Bharosa on Paytm

Trust, turned into growth.

Udhaar made safe, verified and self-collecting, on the Soundbox the merchant already has and the Paytm app the customer already uses.

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SPEAKER NOTES

Open (10 s). "Every kirana in India is secretly a lender. We built Bharosa to make that lending safe, verified and a source of growth, using pieces Paytm already has."
Who leads: business judges → lost sales, new customers, Diwali; technical mentors → the verified receivable and the AI firewall; risk people → consent, caps and who carries which risk.

THE PROBLEM

Every kirana is secretly a lender, and it lends blind.

A customer takes ₹340 of goods today and pays "later". The shop has just made an **unsecured, interest-free loan**, with none of a lender's tools.

No risk check
Decided by gut feel, at a busy counter, in seconds.

No proof
A line in a notebook. The customer never agreed to it.

No collection
Chasing a neighbour for money is awkward, so it doesn't happen.

No funding
Credit is paid for out of the money meant for stock.

▼

Refuses new customers
"I don't know you" means no credit, and no sale.

Caps good customers
Reliable regulars are rationed with everyone else.

Can't stock up
Cash stuck in udhaar means a thin shelf at Diwali.

> **So the shopkeeper rations credit out of fear, and every "no" is a lost sale.**

SPEAKER NOTES

Problem (20 s). "Udhaar is a kirana's loyalty programme and its weapon against quick commerce. But the shopkeeper lends blind: no risk check, no proof, no collections, no funding. So they ration it, and lose sales."
Point at the four red boxes left to right, then the bottom line.

THE CHALLENGE

The problem isn't the notebook. Trust is invisible.

Khata apps already digitised the notebook. They record udhaar; they don't make it **trustworthy to anyone else**. If trust became visible and verifiable, credit would stop being a risk to ration and become a tool to grow with.

Making trust verifiable needs four things at once:

A device at the counter

Where the credit happens, at the moment it happens.

Paytm Soundbox ✓

An app in the customer's hand

So the customer can agree to the debt themselves.

Paytm app ✓

A payment rail

So repayment leaves a digital trail and can be automated.

UPI · UPI AutoPay ✓

Lenders

So verified receivables can fund the shop's growth.

Paytm lending partners ✓

> **Paytm already has all four. Nobody else has them together.**

SPEAKER NOTES

Insight (15 s). "The problem isn't the notebook. It's that trust is invisible. Make it verifiable and credit becomes a growth tool."

Why Paytm (15 s). "Only Paytm has the device on the counter, the app in the customer's hand, the payment rail and the lenders, plus a merchant network that makes portable trust possible."

THE SOLUTION

Bharosa: sell more, get paid, stock up.

Three merchant benefits, built on one trust layer and pulled together by a weekly growth plan.

1 · Sell more, safely

Udhaar Guardrail a safe limit for every request, in a second

Portable Trust new customers bring their Bharosa Score, with consent

Monthly basket regular orders, predicted and offered

2 · Get paid without chasing

Scan-to-Confirm the customer agrees to the debt on their own phone

Smart reminders that understand the reply

Salary-week auto-pay UPI AutoPay, the compliant way

3 · Stock up without cash

Festive plan Diwali demand from last year's sales

Stock Advance a partner lender pays the distributor directly

20% of repayments repay the lender automatically

Growth plan

A weekly to-do list for the merchant, ranked by impact in rupees. One tap to approve each action.

Trust layer

Verified receivables · Bharosa Score · consent · exposure caps · fraud checks

SPEAKER NOTES

Product (40 s), part 1. "At the counter, the Guardrail suggests a safe limit in a second. The customer confirms on their own phone, so the debt is verified. A customer new to the shop can share their Bharosa Score and get credit safely: a new customer won. Reminders understand replies; auto-pay collects in salary week. And for Diwali, Bharosa plans the stock and a partner lender pays the distributor directly, repaid from 20% of incoming repayments."

THE CORE IDEA

One step changes everything: the verified receivable.

A debt both sides acknowledged, with a digital repayment trail, is no longer a note. It is **data**, and data can be used.

TODAY · KHATA APP

Shopkeeper writes "Ramesh ₹200"

▼

A one-sided claim. Could be wrong, inflated or disputed.

▼

Useless to anyone else.

BHAROSA

Shopkeeper records "Ramesh ₹200"

▼

Ramesh confirms on his own phone, in his logged-in Paytm account.

▼

A verified receivable + a UPI repayment trail.

Bharosa Score
A credit identity built from everyday life.

Safe limits
For the next request, here and at other Paytm shops.

Collateral
For a Stock Advance from a partner lender.

SPEAKER NOTES

"Everything in Bharosa flows from this one step. Once a debt is confirmed by the customer, it can decide limits, build a credit identity, and get the shop funded."
Why would a customer confirm? One scan, and on-time repayment of confirmed credit raises their score: higher limits at every Paytm shop.

HOW IT WORKS

One transaction, end to end: Ramesh asks for ₹340.

He already owes ₹1,700 here, 19 days overdue.

Shopkeeper	Soundbox	Bharosa platform	Ramesh's phone
1 Says "Ramesh 340"		**2** Who is "Ramesh"? Rule match; the **Counter Agent** only if the sentence is messy	
		3 Risk engine: score 662 → base ₹3,215 · 19 days overdue → ×0.6 · minus ₹1,700 owed → **₹200**	
	4 Caution. Give ₹200 on credit, collect ₹140 now. Why? shows the reasons		
5 Taps **Use suggestion**		**6** Entry **proposed** · one-time request for Ramesh, 15 minutes	
	7 Shows the request QR		**8** Scans · "Pay ₹140 & confirm"
		9 Is this Ramesh? · ₹140 recorded as a sale · entry **verified** · shared-device check · notify every screen	
	10 "✓ Confirmed by Ramesh", within a second		**11** Sees ₹200 in My dues; score explained

a person acts · code decides · AI may help **AI understood one sentence, at most. Everything after it is deterministic code.**

SPEAKER NOTES

Walk the numbers once, slowly: 662 → ₹3,215 → ×0.6 because he's 19 days overdue → minus the ₹1,700 he owes → ₹200. "The shopkeeper still makes the sale: ₹200 on credit, ₹140 now."

Switch to Live Demo now (tab at the top): Priya 340 (green) → Ramesh 340 (amber) → Use suggestion → scan on Ramesh's phone → confirm. Then Anita 2000 (Portable Trust), Diwali, reminders, basket. Come back to slide 7 for the architecture.

THE ARCHITECTURE

Where everything sits, and where the agents sit.

Soundbox
classic or AI · voice, displays, QR

Paytm for Business
the merchant · approvals, views

Paytm app
the customer · scan, confirm, pay, consent

BHAROSA PLATFORM

1 Access
who you are, what you may see
identity & roles · consent checks · privacy in one place

2 AI layer: the agents
proposes, never decides
Speech · Counter Agent · Recovery Agent · Growth Agent (Munim)

3 Validation: the firewall
every AI output crosses it
strict JSON schema · policy rules · every ₹ and name must exist · else rule / template

4 Trust core
deterministic · the source of truth
ledger & entry states · score · risk: limits, exposure cap · Portable Trust · QR request service

5 Money & growth
deterministic maths
payments · FIFO · 20% split · AutoPay schedule · festive plan · basket · Stock Advance · growth candidates

6 Protection
shared-device check · audit log

7 Events
"changed" pings · spoken lines

Rails: UPI · UPI AutoPay · credit line on UPI (partner banks) · lending partners · distributor payments · licensed credit-information partner

Agents live only in 2
They turn messy language into structured proposals, and numbers into words.

3 is the firewall
Nothing an agent says reaches 4 or 5 without a validator.

4–5 decide
Same input, same output. Exact, auditable, explainable.

Humans approve money
The merchant taps Approve; the customer confirms.

SPEAKER NOTES

For technical judges, lead here. Read the stack top to bottom: Paytm surfaces in, rails out. "The agents sit in one layer only, layer two. Everything they produce crosses a validator, layer three, before deterministic code in layers four and five does anything. Agents can read; they can never write."

Prototype mapping: 2 = app/agents + app/ai/sarvam.py, 3 = the validators in each agent module, 4 = app/core/score, risk, requests, 5 = payments, advance, festive, basket, mandates, growth, 7 = app/bus.py.

WHERE AI STOPS

AI where the world is messy. Code where money moves.

NON-DETERMINISTIC · AI (MAY VARY)

- voice → Speech-to-text → Counter Agent
- customer's reply → Recovery Agent
- growth candidates (code) → Growth Agent writes words
- merchant's question → Munim + read-only tools
- verdict (code) → translate + speak

VALIDATOR

FIREWALL

DETERMINISTIC · CODE (SAME INPUT, SAME OUTPUT)

- → risk engine → **safe limit + reasons**
- → **promise stored** , auto-pay offered
- → **plan text shown** ; the order stays code's
- → **answer shown** , every ₹ from a tool
- → **one spoken line** , same ₹ figures

If a proposal fails the check

Bad JSON, an unknown customer, a ₹ figure not in our data, or an 8-second timeout → the rule or template result is used instead, and logged as such.

Never AI

Score · limits · ledger · payments · the 20% split · Stock Advance sizing · mandates · fraud checks.

SPEAKER NOTES

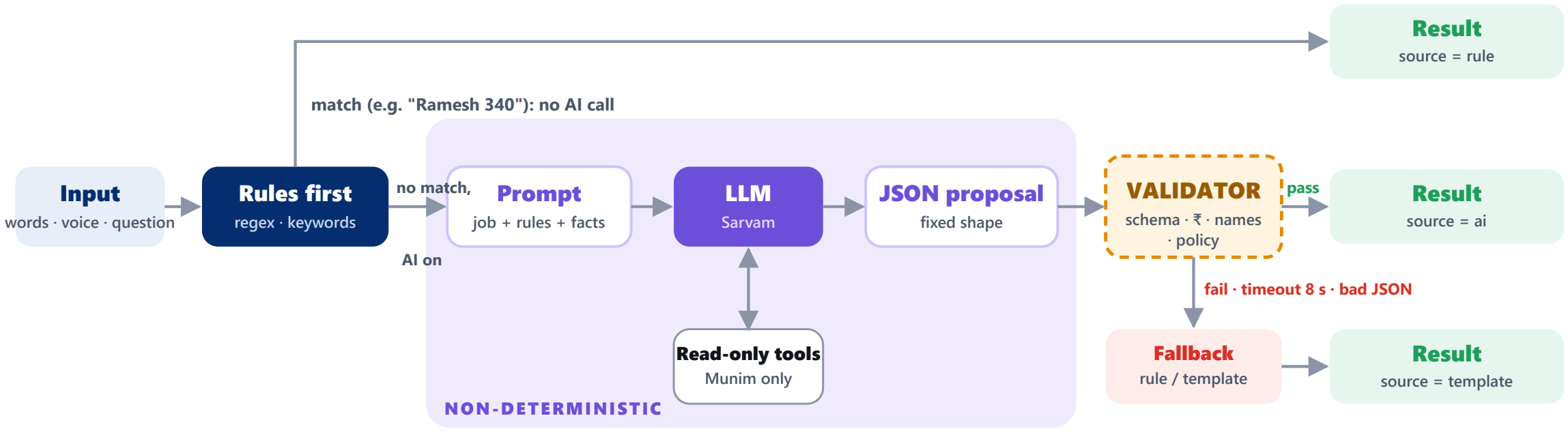
AI (20 s). "AI handles what's messy: Kannada and Hindi speech, conversations, explanations. Code handles money: scores, limits, payments. The AI suggests; it never decides."

Why this split: LLMs are good at messy language, many languages and code-mixing, polite writing and explanations. They are bad at exact arithmetic, consistency, auditability and not inventing facts. So we use them only where their strengths matter.

WHAT "AGENT" MEANS HERE

An agent = LLM + one job + a validator + a fallback.

one job · optional read-only tools · strict JSON output · validator · fallback Rules always run first; the LLM is only called when they can't answer.



Agents read, never write

Tools only fetch. Any change goes through deterministic services, usually after a human taps.

Every result is labelled

The activity log shows rule, ai or template for each decision.

AI off? Still works

With the switch off, every flow runs on rules and templates. A test makes sure no AI call is made.

SPEAKER NOTES

"In Bharosa an agent is a narrow thing: an LLM with one job, a strict JSON format, a validator and a fallback. Simple inputs like 'Ramesh 340' never touch the LLM. The LLM only proposes; the validator decides; if anything goes wrong, the rule or template answer is used and the product keeps working."
If asked: the demo runs with AI off by default (SARVAM_ENABLED=false); test_ai_off.py asserts zero outbound calls.

THE AGENTS, AND WHY EACH ONE

Four agents, each where rules alone fall short.

Speech

Job: voice → text; one short Kannada line back

Why AI: speech recognition is inherently AI

Never: decides anything

Checks: spoken line keeps the same ₹ figures

Counter Agent

Job: "Ramesh ne 340 ka saaman liya" → {customer, amount}

Why AI: many languages, code-mixing, number words

Never: invents a customer or amount; sets a limit

Checks: customer in the list · amount ₹1–₹50,000

Recovery Agent

Job: polite reminders; "salary 1 tareekh ko" → promise ₹ on 1 Nov

Why AI: replies are open-ended; templates can't read them

Never: changes dues; accepts a promise outside policy

Checks: only ₹ = dues · promise ≤ dues, within 30 days

Growth Agent (Munim)

Job: words the weekly plan; answers "How do I pay Sharma on Friday?"

Why AI: turns several data sources into plain advice

Never: changes a number or the ranking; acts without a tap

Checks: same actions, code's order · every ₹ from the facts

Why not more agents?

Each one adds non-determinism, latency and cost. Scores, limits, payments, the split, advance sizing, auto-pay and fraud checks are exact maths: they stay code.

Why not zero agents?

Then the shopkeeper types exact formats at a busy counter, reminders are generic and replies are ignored. That's just another khata app.

SPEAKER NOTES

"We didn't add agents for the sake of it. There are exactly four, and each one sits where the input is genuinely messy human language. Look at the bottom of every card: what it can never do. And the amber row: the check its output must pass."
Judge question "why agents at all?": only where the input is messy human language; each output is validated by code, with a fallback.

THE FLOW

Where each agent works along the journey.

Six steps from the counter to growth. AI appears at the edges, where people speak; code holds the middle, where money moves.

	Ask at the counter	**Check** safe limit	**Decide** the offer	**Confirm** on the phone	**Collect** get paid	**Grow** weekly plan
Who acts	Shopkeeper speaks or types	Code	Code; shopkeeper accepts or overrides	Customer scans and confirms	Customer replies, pays, sets up auto-pay	Merchant approves
Agent	**Speech → Counter Agent (only if rules fail)**	none	**Speech says the verdict**	**Speech says "confirmed"**	**Recovery Agent: writes reminders, reads replies**	**Growth Agent: words the plan; Munim answers**
Code that decides	Rule parser · validator	Score · overdue penalty · exposure cap · Portable Trust	Request service · ledger (proposed)	Identity check · sale · entry verified · device check	Promise · AutoPay schedule · FIFO · 20% split	Festive plan · Stock Advance · basket · ranking

Counter → Confirm

One QR request ties the counter to the right customer's phone. Any Soundbox works: printed QR, AI display or the merchant's phone.

Collect

"Salary 1 tareekh ko aati hai" → a promise on 1 Nov → salary-week auto-pay, with a notice the day before.

Grow

Code ranks Diwali stock, overdue reminders and the monthly basket by rupee impact; the agent only explains them.

SPEAKER NOTES

Tie this back to what they just saw in the Live Demo: Ask/Check/Decide/Confirm was the Soundbox and Ramesh's phone; Collect was the reminder and the reply; Grow was the plan on the merchant home. "Every violet cell is optional: switch AI off and the same journey runs on rules and templates."

CONTAINING NON-DETERMINISM

Six controls keep the AI honest.

1
Strict output formats
The LLM must return JSON of a fixed shape. Anything else is rejected.

2
Grounding checks
Every ₹ figure and every customer the AI names must already exist in our data.

3
Rules first, AI second
"Ramesh 340" never reaches the LLM. Faster, cheaper, predictable.

4
Humans approve money
The merchant taps Approve or Accept; the customer confirms every credit.

5
Fallbacks everywhere
AI down, slow or wrong? Templates and rules take over. The product keeps working.

6
In production, measure
Phrase test sets per language, accuracy monitoring, a log of every proposal and verdict.

> **"We use AI where the world is messy (speech, languages, conversations, explanations) and code where money moves. The AI can suggest; it can never decide or pay."**

SPEAKER NOTES

This is the one-sentence answer for judges; say the quote verbatim. If asked about hallucinated numbers: control 2. If asked about the AI being down on stage: control 5, and the demo already runs with AI off.

PROTOTYPE VS PRODUCTION

The intelligence is real. The rails are adapters.

Part	In the real solution	In our prototype
Soundbox	Paytm devices send voice and text	A web page that behaves like the AI Soundbox
Customer and merchant apps	Bharosa inside the Paytm app and Paytm for Business	Phone web apps styled like them, with phone + OTP login
Ledger, score, limits, requests, split, advance sizing, fraud check	**Same logic**	**Real code**, covered by tests
UPI, AutoPay, credit line, lender, distributor	Real rails and partners	Simulated adapters with the same shape
AI	Sarvam (or Paytm's own models), with evaluation sets	Sarvam behind an on/off switch; off by default
Infrastructure	Paytm cloud: Postgres, event streaming, caches	FastAPI + SQLite on a laptop; a secure tunnel for real phones

> **Swap the adapters, and this runs on Paytm.**

SPEAKER NOTES

"The intelligence is real; the rails are simulated behind adapters shaped like Paytm's. Swap the adapters, and this runs on Paytm."

WHO BENEFITS

Good for the merchant, good for Paytm, no balance-sheet risk.

The merchant

- **More customers:** serve people they've never met
- **More per customer:** clear limits, monthly baskets
- **Seasonal growth:** Diwali stock planned and funded
- **Money back faster:** confirmed debts, smart reminders, auto-pay
- **Fewer losses, no awkwardness**

Paytm

- **Soundbox engagement** beyond payment alerts
- **Lending distribution** via Stock Advances
- **New consumers** install the app to confirm and build a score
- **More UPI and AutoPay volume**
- **A network moat:** Portable Trust grows with every merchant

Customers and lenders

- **Customers:** credit at more shops and a credit identity from everyday life
- **Lenders:** short, self-repaying advances backed by verified receivables, and new pre-verified borrowers

No balance-sheet risk for Paytm

Paytm doesn't lend udhaar. The shop keeps its udhaar risk, now capped and informed; regulated partners carry advances and above-limit credit.

SPEAKER NOTES

Close the business case. "Bharosa grows merchants, grows Paytm's lending and app usage, and carries no balance-sheet risk." Revenue lines if asked: lending distribution fees on advances and loans, bank fees for credit-line customers acquired at the counter, collections, an optional premium tier.

RISKS, DESIGNED OUT

Key design decisions, and what we rejected.

We chose	Instead of	Because
Customer confirms on their own phone	Shopkeeper-only records	A one-sided record can't be trusted by anyone else
Any Soundbox + a printed QR	Requiring the AI Soundbox	Most merchants have speaker-only devices: no new hardware
A transparent score formula	A black-box ML score	Customers, merchants and regulators must understand it
Suggested limit, override allowed	A hard block	It's the shopkeeper's money and relationship
Consent before using the network score	Sharing by default	Privacy, and likely regulation
Merchant sees a band + limit only	Full history	Minimal disclosure: no neighbourhood gossip
Exposure cap across shops	Per-shop limits only	Stops one customer borrowing everywhere at once
Lender pays the distributor directly	A cash loan to the merchant	Can't be misspent; lower risk; cheaper
Repay from 20% of repayments	Fixed instalments	Follows the shop's cash flow
Merchant-triggered AutoPay, notice the day before	"Debit when salary lands"	That's how UPI AutoPay actually works
AI never touches money	AI decides limits	Non-determinism + money = unacceptable

SPEAKER NOTES

For risk and compliance people, lead here. Pick two or three rows depending on who is asking: consent and minimal disclosure, the exposure cap, the lender paying the distributor directly, and AutoPay compliance.

The questions we expect, and how the design answers them.

1 Isn't this just Khatabook?

Khatabook records. We check risk before credit, get the customer's confirmation, win new customers through the network, collect automatically and fund stock.

Built in: Guardrail · Scan-to-Confirm · Portable Trust · Stock Advance

2 Why would a customer confirm their own debt?

One scan, and on-time repayment of confirmed credit raises their score: higher limits at every Paytm shop.

Built in: score breakdown shown to the customer

3 Is AI deciding who gets credit?

No. A transparent formula decides; the shopkeeper has the final say. AI handles language only.

Built in: validator firewall · "Why?" reasons · override

4 Why agents at all?

Only where the input is messy human language. Each agent's output is validated by code, with a fallback if it fails.

Built in: rules first · AI switch · source on every log line

5 Who takes the loss if a customer defaults?

The shopkeeper, as today, but with far better information and a capped limit. Paytm takes no credit risk; advances come from regulated lenders, backed by verified receivables and repaid from 20% of collections.

Built in: overdue penalty · exposure cap · lender split

SPEAKER NOTES

Answer in one breath each; point at the green "built in" line, which is what they can see in the demo. If a judge asks something here live, jump to this slide (press End, then ←).

Regulation, fraud, hardware and money.

6 Isn't sharing scores regulated?

Likely. So sharing is customer-controlled and revocable, disclosure is minimal, and production scoring would run through a licensed partner.

Built in: Share / Not now · Revoke · band + limit only

7 Can auto-pay debit on salary day?

Not instantly. UPI AutoPay needs notice the day before, so we collect on a date the customer picks in salary week, and they can skip.

Built in: cap · collection day · pre-debit notice

8 Can shops fake receivables to get advances?

We look for shared devices, instant confirmations, round amounts and money going in circles, and exclude suspicious entries from advance eligibility.

Built in: shared-device flag · verified-only eligibility

9 Does it need the new AI Soundbox?

No. Any Soundbox and the printed QR work; the AI Soundbox just adds voice and screens.

Built in: the QR request service works from any device

10 How does Paytm make money?

Lending distribution fees on advances and loans, bank fees for credit-line customers acquired at the counter, collections, and an optional premium tier, with no balance-sheet risk.

Built in: Stock Advance · credit line on UPI at the counter · AutoPay

SPEAKER NOTES

Fraud-ring follow-up: "A dishonest shop could invent customers and confirm from one phone. A confirmation from a device already used by another customer at the same shop is flagged and kept out of advance eligibility."

Bharosa **on Paytm**

Make trust verifiable, and credit becomes a growth engine.

AI handles the language. Code handles the money. Paytm already has every piece.

Sell more, safely
Guardrail · Portable Trust · basket

Get paid without chasing
Scan-to-Confirm · reminders · auto-pay

Stock up without cash
Diwali plan · Stock Advance

Bharosa: trust, turned into growth.

SPEAKER NOTES

Close (10 s). "Bharosa grows merchants, grows Paytm's lending and app usage, and carries no balance-sheet risk. Trust, turned into growth."
Then offer the phone: "Scan this and try it yourself" (login QRs are on the Live Demo strip).